MPG No. 23.5.86 $6'/m'$ [Type III, hexagonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & T_{xxzx} & 0 \\ 0 & 0 & 0 & -T_{xxyz} & -T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyz} & -T_{xxyz} & 0 & 0 & 0 & -T_{xxzx} \\ T_{xxzx} & -T_{xxzx} & 0 & 0 & 0 & T_{xxyz} \\ 0 & 0 & 0 & -T_{xxzx} & T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = \mathbb{T}^{(1)}_{4s}$

$T_{xxzx} = \mathbb{T}^{(1)}_{4c}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & T_{xxzx} & 0 \\ 0 & 0 & 0 & -T_{xxyz} & -T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxyz} & T_{xxyz} & 0 & 0 & 0 & T_{xxzx} \\ -T_{xxzx} & T_{xxzx} & 0 & 0 & 0 & -T_{xxyz} \\ 0 & 0 & 0 & -T_{xxzx} & T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = 2\mathbb{M}^{(1)}_{3c}$

$T_{xxzx} = -2\mathbb{M}^{(1)}_{3s}$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & T_{xxzx} & 0 \\ -T_{xxyz} & -T_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{xxzx} & -T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = \mathbb{M}^{(2)}_{3c}$

$T_{xxzx} = \mathbb{M}^{(2)}_{3s}$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & T_{yzxy} \\ T_{yzxy} & -T_{yzxy} & 0 & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$T_{yzxx} = \mathbb{M}^{(3)}_{3c}$

$T_{yzxy} = \mathbb{M}^{(3)}_{3s}$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & T_{xxzx} & 0 & T_{xxzy} & T_{xxxz} & 0 \\ 0 & 0 & 0 & -T_{xxyz} & -T_{xxzx} & 0 & -T_{xxzy} & -T_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzxy} \\ -T_{yzxy} & T_{yzxy} & 0 & 0 & 0 & T_{yzxx} & 0 & 0 & T_{yzxx} \\ 0 & 0 & 0 & -T_{xxxz} & T_{xxzy} & 0 & -T_{xxzx} & T_{xxyz} & 0 \end{bmatrix}$$

$$
\begin{aligned}
T_{xxyz} &= 2\mathbb{M}_{3c}^{(1)} + \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)} \\
T_{xxzx} &= -2\mathbb{M}_{3s}^{(1)} + \mathbb{M}_{3s}^{(2)} + \mathbb{T}_{4c}^{(1)} \\
T_{xxzy} &= 2\mathbb{M}_{3c}^{(1)} - \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)} \\
T_{xxxz} &= -2\mathbb{M}_{3s}^{(1)} - \mathbb{M}_{3s}^{(2)} + \mathbb{T}_{4c}^{(1)} \\
T_{yzxx} &= -2\mathbb{M}_{3c}^{(1)} + \mathbb{T}_{4s}^{(1)} \\
T_{yzxy} &= -2\mathbb{M}_{3s}^{(1)} - \mathbb{T}_{4c}^{(1)}
\end{aligned}
$$

* Rank 4 tensor (a).

$$
\begin{bmatrix}
T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzxy} \\
T_{yzxy} & -T_{yzxy} & 0 & 0 & 0 & -T_{yzxx} & 0 & 0 & -T_{yzxx} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxx} &= \mathbb{M}_{3c}^{(3)} \\
T_{yzxy} &= \mathbb{M}_{3s}^{(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & 0 & T_{xxxz} \\
0 & 0 & T_{yyyz} \\
0 & 0 & 0 \\
-T_{xxxz} & T_{yyyz} & 0 \\
0 & 0 & 0 \\
0 & 0 & -T_{yyyz} \\
0 & 0 & 0 \\
T_{xxxz} & -T_{yyyz} & 0 \\
0 & 0 & -T_{xxxz} \\
-T_{yyyz} & -T_{xxxz} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxxz} &= \mathbb{T}_{4c}^{(1)} \\
T_{yyyz} &= -\mathbb{T}_{4s}^{(1)}
\end{aligned}
$$